

10.6.13 Lab - Research Password Recovery Procedures

Step 1: Describe the purpose of the configuration register.

What is the purpose of the configuration register?

ANSWER: The configuration register can be used to change the way the router boots, options for booting, and console speed.

What command changes the configuration register in global configuration mode?

ANSWER: config-register

What command changes the configuration register in ROMMON mode?

ANSWER: The configuration register can be used to change the way the router boots, options for booting, and console speed.

Step 2: Determine configuration register values and their meanings.

Research and list the router behavior for the following configuration register values.

0x2102

ANSWER: For the configuration register value 0x2102, a router will load the IOS from flash memory and then load the start-up configuration from the NVRAM, if present. If no operating system is found, the router will boot to ROMMON.

0x2142

ANSWER: For the configuration register value 0x2142, a router will load the IOS from flash memory, ignore the start-up configuration in NVRAM, and provide a prompt for initial configuration dialog. If no operating system is found, the router will boot to ROMMON.

What is the difference between these two configuration register values?

ANSWER: The 0x2102 setting is for normal router operation. The 0x2142 setting bypasses the start-up configuration allowing a user to recover or reset the enable password.

Part 2: Document the Password Recovery Procedure for a Specific Cisco Router

For Part 2, you will describe the exact procedure for recovering or resetting a password from a 2900 series Cisco router and answer questions based on your research.

Step 1: Detail the process to recover a password on a specific Cisco router.

Research and list the steps and commands to recover or reset the enable or enable secret password from your Cisco router. Summarize the steps in your own words.

ANSWER:

To recover or reset the enable password on the Cisco 2900 series router, complete the following steps:

1. Establish a terminal connection with the router using Tera Term or other terminal emulator.
2. Boot to ROMMON by either removing flash and rebooting or selecting Alt-b during a reboot.
3. Type confreg 0x2142 at the rommon prompt.
4. Type reset at the next rommon prompt.
5. Type no at the initial configuration dialog.
6. Type enables at the router prompt.
7. Type copy startup-config running-config to load the startup configuration.
8. Type configure terminal.
9. Record an unencrypted enable password. Reset an encrypted enable password.
10. **In configuration mode, type config-register 0x2102.**
11. In privileged EXEC mode, type copy running-config startup-config to save configuration.
12. Use the show version command to verify configuration register settings.

Step 3: Answer questions about the password recovery procedure.

Using the process for password recovery, answer the following questions.

Describe how to find the current setting for your configuration register.

ANSWER: The show version command will provide the current setting for the configuration register.

Describe the process for entering ROMMON mode.

ANSWER: A user can remove the flash memory and restart the router to boot to the ROMMON utility. A user can also boot the router and select alt+b when using Tera Term on real equipment.

What commands do you need to enter the ROMMON interface?

ANSWER: (2900-series) A user would need to enter `confreg 0x2142` to change the configuration setting, followed by `reset` to restart the router.

What message would you expect to see when the router boots?

ANSWER: If a router does not load the startup configuration, a user would expect to see the message “Continue with configuration dialog?”

Why is it important to load the startup configuration into the running configuration?

ANSWER: Loading the startup configuration into the running configuration ensures that the original startup configuration remains intact if the user saves during the password recovery process.

Why is it important to change the configuration register back to the original value after recovering password?

ANSWER: Returning the configuration register to the original value will ensure that the router will load the startup configuration during the next reload.

Reflection Question

Why is it of critical importance that a router be physically secured to prevent unauthorized access?

Because the password recovery procedure can only be performed using a console connection, which requires direct physical access to the device, preventing unauthorized users access to the physical device is an imperative part of an overall security plan.